visStatistics: A transparent visual decision pipeline for statistical hypothesis tests in R

by Sabine Schilling

Abstract We present **visStatistics**, an R package that implements a transparent decision pipeline for selecting, executing, and visualising common statistical hypothesis tests between two vectors. Given vectors of class `numeric`, `integer`, or `factor`, the main function `visstat()` routes the analysis according to variable types, sample size, and displayed assumption diagnostics. Visual outputs — box plots, bar charts, regression lines with confidence bands, mosaic plots, and residual diagnostics — are annotated with test statistics, assumption-check results, and post-hoc comparisons where applicable. The decision logic follows the general linear model framework, assessing normality on pooled model residuals rather than per group, which preserves statistical power. The package is particularly suited for server-side R applications where end users interact exclusively via a web interface, and for time-constrained settings such as statistical consulting or introductory teaching. All tests are applied using their default settings from base R, keeping the dependency footprint minimal and the decision tree reproducible.

1 Introduction

Most routine data analyses reduce to a comparatively small set of inferential frameworks: group comparisons, contingency-table analyses, correlation, and simple regression (Sato et al., 2017; Chicco et al., 2025). Selecting the appropriate test within each framework, however, requires evaluating variable types, sample sizes, and distributional assumptions — a process that is error-prone, time-consuming, and a recurring bottleneck in statistical consulting and undergraduate teaching.

Several R packages address parts of this problem. **compareGroups** (Subirana et al., 2014) automates test choice via a per-group Shapiro–Wilk normality test and produces publication-ready descriptive tables, but its primary output is tabular rather than graphical, and assumption diagnostics are not displayed alongside results. **ggstatsplot** (Patil, 2021) produces information-rich **ggplot2**-based (Wickham, 2016) plots with embedded test results, but requires the user to select the appropriate plotting function for each data-type combination (e.g., `ggbetweenstats()` for group comparisons, `ggscatterstats()` for regression). **rstatix** (Kassambara, 2025) provides a pipe-friendly framework for common tests with tidy output, and **ggpubr** (Kassambara, 2026) adds statistical annotations to publication-ready plots; both leave test selection entirely to the user.

visStatistics takes a different approach: a single function `visstat(x, y)` accepts any combination of numeric and categorical vectors and resolves the full decision pipeline — test selection, execution, assumption checking, and visualisation — without any further input. Three design choices distinguish it from the packages above.

The contribution is not a new hypothesis test, but a reproducible software workflow around existing tests. `visstat()` combines a unified two-vector interface, explicit routing by input class and diagnostics, graphical assumption checks shown alongside the selected result, and branch-specific post-hoc procedures. The aim is to make routine default analyses transparent and inspectable, not to replace statistical judgement in cases where the displayed diagnostics suggest that another model is more appropriate.

First, normality is assessed on the pooled residuals of a linear model `lm(y ~ x)`, following the general linear model (GLM) framework (Searle, 1971), rather than per group. Group-wise testing is statistically less powerful and more likely to reject normality unnecessarily, pushing the analysis to non-parametric methods (Lumley et al., 2002). When heteroscedasticity is subsequently detected, group-wise normality diagnostics are shown additionally, since Welch's methods use group-specific variance estimates.

Second, assumption diagnostics are always displayed, regardless of whether assumptions are met. This enables users to visually override the automated decision in individual cases, and supports the pedagogical goal of making assumption checking an integral part of the analysis rather than an afterthought.

Third, all tests use their default settings from base R functions, keeping the dependency footprint minimal and the decision tree transparent. The package does not wrap external statistical engines; the automated layer is purely in the routing logic.

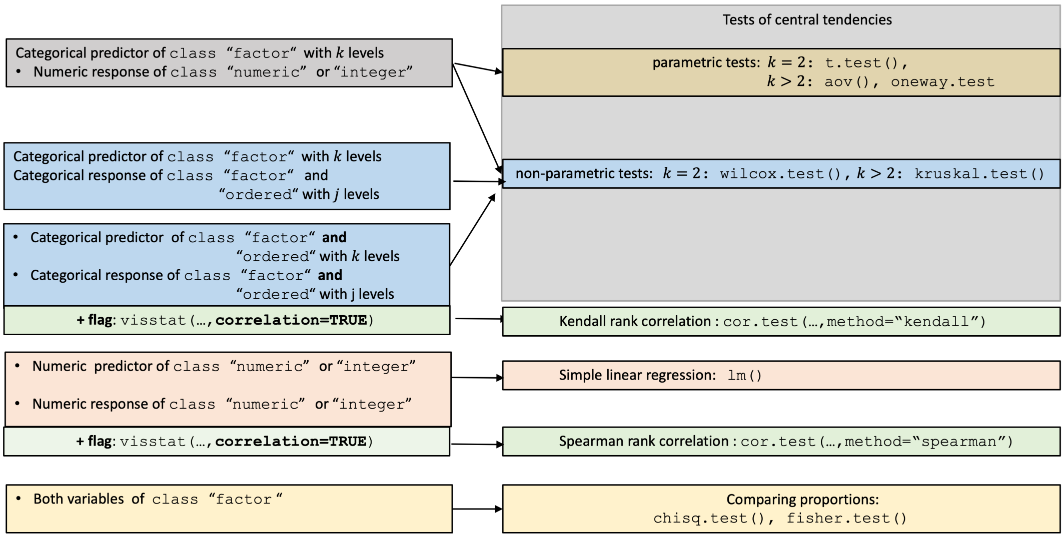

Figure 1: Overview of all implemented tests by input class.

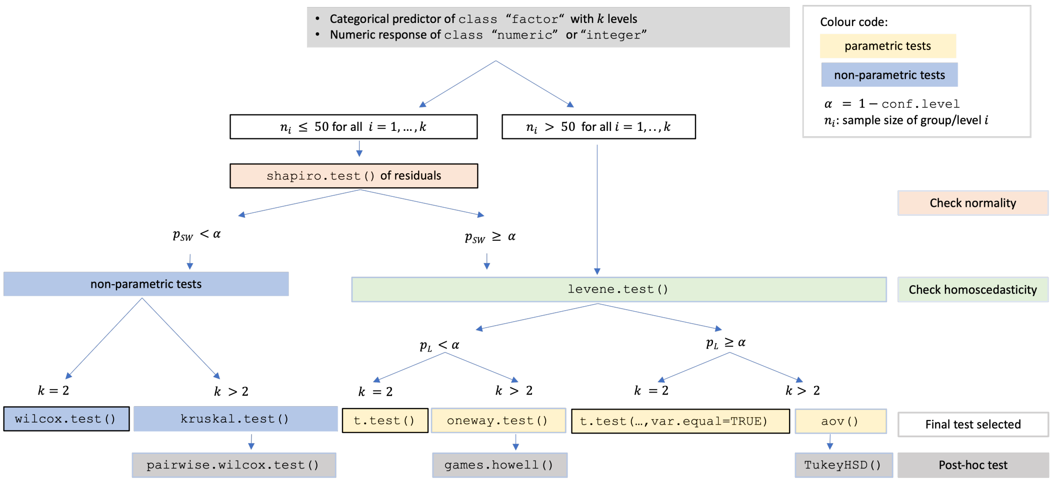

Figure 2: Decision tree for a numeric response and categorical predictor, based on sample size, normality of residuals, and homoscedasticity.

2 Package overview

visStatistics is available on CRAN. The package accepts input in two equivalent forms:

```
# Recommended form:
visstat(x, y)

# Formula interface:
visstat(y ~ x, data = dataframe)
```

where `x` and `y` are vectors of class `"numeric"`, `"integer"`, or `"factor"`. The function returns an object of class `"visstat"` with `print()`, `summary()`, and `plot()` S3 methods, and optionally writes graphics to disk via **Cairo** (Urbanek and Horner, 2025).

3 Decision logic

The significance level $\alpha = 1 - \texttt{conf.level}$ is user-controlled (default 0.95). Figure 1 gives a high-level overview of all implemented tests by input class; Figure 2 details the decision path for a numeric response and categorical predictor. Technical definitions of all test statistics are collected in Appendix B.

Figure 1 is read by input class. Numeric responses with categorical predictors enter the central-tendency branch; two unordered factors enter the proportion-comparison branch; two numeric variables enter simple linear regression unless `correlation = TRUE`, which switches to Spearman rank correlation; and two ordered factors can be analysed as Kendall rank correlation when `correlation = TRUE`. Figure 2 then expands the central-tendency branch. Following the linear-model framing introduced above and detailed in Appendix A, the first split is standardised residual normality. If Shapiro–Wilk rejects residual normality, `visstat()` selects the non-parametric path (`wilcox.test()` for two groups, `kruskal.test()` for more than two groups, with pairwise Wilcoxon post-hoc comparisons. If residual normality is not rejected, Levene–Brown–Forsythe decides between the equal-variance parametric path (`t.test(var.equal = TRUE)` or `aov()` with `TukeyHSD()`) and the heteroscedastic Welch path (`t.test()` or `oneway.test()` with `games.howell()`).

3.1 Numeric response and categorical predictor: comparing central tendencies

A linear model `lm(y ~ x)` is fitted and assumption diagnostics are always displayed via `vis_lm_assumptions()` (see Appendix A). The Shapiro–Wilk test (Shapiro and Wilk, 1965) is applied to internally studentised residuals from `rstandard()` (Cook and Weisberg, 1982). When all groups contain more than 50 observations, normality is assumed by the central limit theorem (Lumley et al., 2002; Rasch et al., 2011).

If normality is rejected ($p_{\text{SW}} \leq \alpha$), non-parametric tests are selected: `wilcox.test()` (Mann and Whitney, 1947) for two groups, or `kruskal.test()` (Kruskal and Wallis, 1952) followed by Holm-adjusted `pairwise.wilcox.test()` (Holm, 1979) for more than two groups.

If normality is not rejected, the Levene–Brown–Forsythe test `levene.test()` (Brown and Forsythe, 1974) (see Appendix B) assesses variance homogeneity. For homoscedastic data, Student's `t.test(var.equal = TRUE)` or Fisher's `aov()` (Fisher and Yates, 1990) with `TukeyHSD()` (Hochberg and Tamhane, 1987) is applied. For heteroscedastic data, Welch's `t.test()` (Welch, 1947; Satterthwaite, 1946) or `oneway.test()` (Welch, 1951) with `games.howell()` (Games and Howell, 1976) is applied. When switching to Welch methods, group-wise normality diagnostics are additionally displayed via `vis_group_normality()`, since the pooled `lm()` residuals are no longer the appropriate diagnostic under heteroscedasticity.

3.2 Both variables numeric: regression or Spearman rank correlation

By default, `visstat()` fits a simple linear regression via `lm()`, regardless of whether GLM assumptions are met, and always displays assumption diagnostics. When `correlation = TRUE` is set explicitly, Spearman's ρ is computed via `cor.test(..., method = "spearman")`. Because the choice between modelling a directional relationship and measuring a monotone association cannot be automated from data types alone, Spearman correlation is never triggered automatically.

3.3 Both variables categorical: comparing proportions

When both variables are categorical, `visstat()` tests independence using Pearson's χ^2 test (`chisq.test()`) or Fisher's exact test (`fisher.test()`), depending on expected cell counts following Cochran's rule (Cochran, 1954): the χ^2 approximation is used if no expected cell count is less than 1 and no more than 20% of cells have expected counts below 5. Yates' continuity correction (Yates, 1934) is applied by default to 2×2 tables. For general $R \times C$ tables, Pearson's χ^2 result is supplemented with a mosaic plot from **vcd** (Meyer et al., 2006, 2024) with tiles coloured by Pearson residuals.

3.4 Ordered factors

When the response is an ordered factor, `visstat()` converts it to integer level codes and follows the non-parametric path. When both variables are ordered and `correlation = TRUE`, Kendall's τ_b (Kendall, 1945; Agresti, 2010) is computed instead, as it corrects for ties explicitly — unavoidable with few ordinal levels (Xu et al., 2013).

4 Examples

The examples follow the same order as the decision logic. The number of plots returned by `visstat()` depends on the input types and the outcome of assumption checks. Most tests produce two panels: an assumption-diagnostic panel and a result panel. When heteroscedasticity is detected and `visstat()`

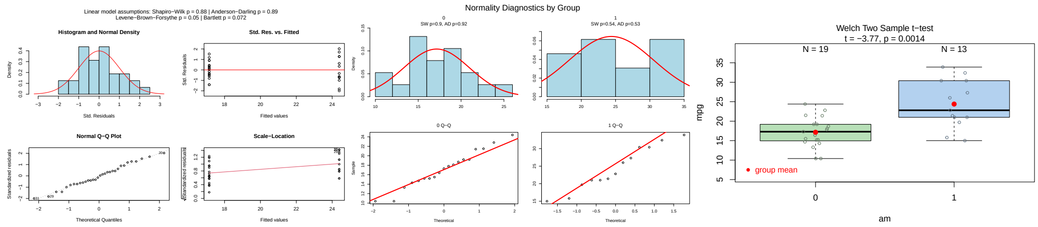

Figure 3: Welch's t-test applied to the `mtcars` dataset (`am` vs. `mpg`). Pooled-residual assumption diagnostics (Shapiro–Wilk, Bartlett, Levene), group-wise normality diagnostics, and box plots of fuel efficiency by transmission type with the Welch t-test result.

routes to a Welch method, a third panel with group-wise normality diagnostics is added, since pooled-residual diagnostics are no longer appropriate under unequal variances. Tests that have no separate assumption diagnostics, such as Kendall's τ_b for two ordered factors, produce a single plot.

The datasets were chosen to exercise the main branches of the decision tree: normal and homoscedastic residuals leading to Student's t-test or one-way ANOVA; normal residuals with unequal variances leading to Welch's t-test or Welch's one-way test; non-normal residuals leading to Wilcoxon or Kruskal–Wallis tests, with Holm-adjusted pairwise Wilcoxon tests for the multi-group extension; numeric–numeric regression and rank correlation; categorical independence tests under and beyond Cochran's rule; and ordinal rank correlation.

4.1 Numeric response and categorical predictor: two groups

For two groups, `visstat()` first evaluates the linear-model assumptions on pooled residuals. If Shapiro–Wilk rejects normality, the decision path switches to the Wilcoxon rank-sum test. If normality is not rejected, the Levene–Brown–Forsythe test decides between Student's equal-variance t-test and Welch's unequal-variance t-test.

Welch's t-test

The *Motor Trend Car Road Tests* dataset (`mtcars`) contains 32 observations, where `mpg` denotes miles per (US) gallon and `am` represents the transmission type (`0` = automatic, `1` = manual). With binary factor `am` and continuous response `mpg`, Shapiro–Wilk does not reject normality of the pooled residuals, while the Levene–Brown–Forsythe test detects heteroscedasticity. The routing therefore leads to Welch's t-test rather than Student's t-test.

```
mtcars$am <- as.factor(mtcars$am)
t_test_stats <- visstat(mtcars$am, mtcars$mpg)
```

Wilcoxon rank-sum test

The Wilcoxon branch is exemplified on examination grades for a class of 44 students (21 girls, 23 boys). After fitting `lm(grade ~ sex)`, Shapiro–Wilk rejects normality of the pooled residuals, so `visstat()` switches to the non-parametric Wilcoxon rank-sum test.

```
grades_gender <- data.frame(
  sex   = as.factor(c(rep("girl", 21), rep("boy", 23))),
  grade = c(19.3, 18.1, 15.2, 18.3,  7.9,  6.2, 19.4, 20.3,  9.3, 11.3,
            18.2, 17.5, 10.2, 20.1, 13.3, 17.2, 15.1, 16.2, 17.0, 16.5,
             5.1, 15.3, 17.1, 14.8, 15.4, 14.4,  7.5, 15.5,  6.0, 17.4,
             7.3, 14.3, 13.5,  8.0, 19.5, 13.4, 17.9, 17.7, 16.4, 15.6,
            17.3, 19.9,  4.4,  2.1)
)
wilcoxon_stats <- visstat(grades_gender$sex, grades_gender$grade)
```

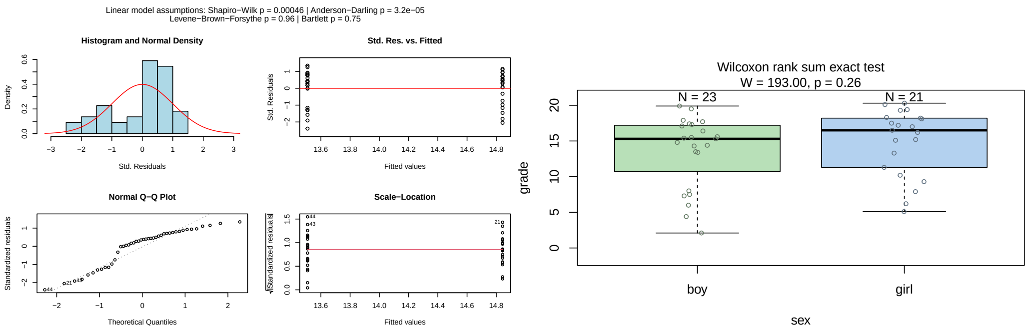

Figure 4: Wilcoxon rank-sum test applied to examination grades by sex. Assumption diagnostics (Shapiro–Wilk rejected; non-parametric path selected) and box plots with the Wilcoxon test result.

4.2 Numeric response and categorical predictor: more than two groups

For more than two groups, the same residual-normality decision is applied first. If residual normality is rejected, `visstat()` applies `kruskal.test()`. Otherwise, `levene.test()` chooses between Fisher's one-way ANOVA (`aov()`) and Welch's heteroscedastic one-way ANOVA (`oneway.test()`). The graphical output consists of two panels. The first panel displays standardised residuals versus fitted values and a normal Q–Q plot; no more than about 5% of the standardised residuals should exceed $|2|$ when residuals are normally distributed, and the Q–Q points should approximately follow the red reference line. The title reports the Shapiro–Wilk and Anderson–Darling normality tests together with Bartlett's and Levene–Brown–Forsythe tests for homoscedasticity. The second panel shows the selected omnibus test as box plots with sample sizes above each group; for the parametric branches, the title also reports the corresponding F statistic. When the largest group has no more than 50 observations, jittered points are overlaid to show the raw data.

ANOVA, Welch ANOVA, and Kruskal–Wallis are omnibus tests: a significant result tells us that *some* group differs, but not which. To identify the differing pairs, `visstat()` tests all pairwise comparisons among the factor levels, defining a family of tests (Abdi, 2007). Because the three omnibus tests rest on different assumptions, each branch uses a matching post-hoc procedure: `TukeyHSD()` after `aov()`, `games.howell()` after `oneway.test()`, and `pairwise.wilcox.test(p.adjust.method = "holm")` after `kruskal.test()`. `TukeyHSD()` controls the family-wise error rate through the studentised range distribution under a common-variance assumption. `games.howell()` uses separate variance estimates and Welch-adjusted degrees of freedom for each pair, making it the post-hoc procedure for the heteroscedastic Welch branch. The Kruskal–Wallis branch uses Holm's step-down adjustment for the pairwise Wilcoxon tests. Pairs whose adjusted post-hoc p-value falls below α are marked with different green letters below the box plots; pairs sharing a letter are not significantly different. The returned object contains the corresponding adjusted p-values.

One-way ANOVA with Tukey HSD

The `npk` dataset reports the yield of peas from an agricultural experiment conducted on six blocks. Shapiro–Wilk does not reject normality of the pooled residuals and the Levene test does not reject homoscedasticity, so `visstat()` applies Fisher's one-way ANOVA followed by Tukey HSD post-hoc comparisons. The omnibus F-test is not significant at $\alpha = 0.05$, and all groups share the same compact-letter label.

```
anova_npk <- visstat(npk$block, npk$yield, conf.level = 0.95)
```

Kruskal-Wallis rank sum test

The `iris` dataset contains petal-width measurements for three species. Here, the residual plots and very small Shapiro–Wilk and Anderson–Darling p-values indicate non-normal residuals. Since Shapiro–Wilk falls below α, `visstat()` switches to `kruskal.test()` followed by Holm-adjusted `pairwise.wilcox.test()`. All three species differ significantly in petal width, as indicated by distinct compact-letter labels.

```
kruskal_iris <- visstat(iris$Species, iris$Petal.Width)
```

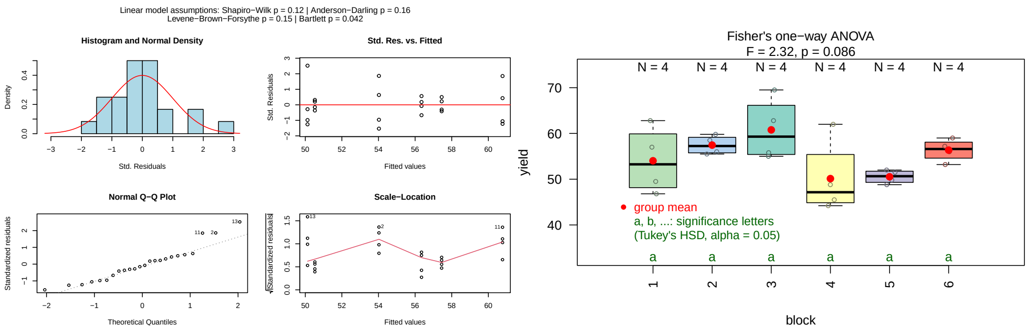

Figure 5: Fisher's one-way ANOVA applied to the `npk` dataset (`block` vs. `yield`). Assumption diagnostics and box plots of pea yield by experimental block; compact letter display (TukeyHSD, $\alpha = 0.05$) shows no significant pairwise differences.

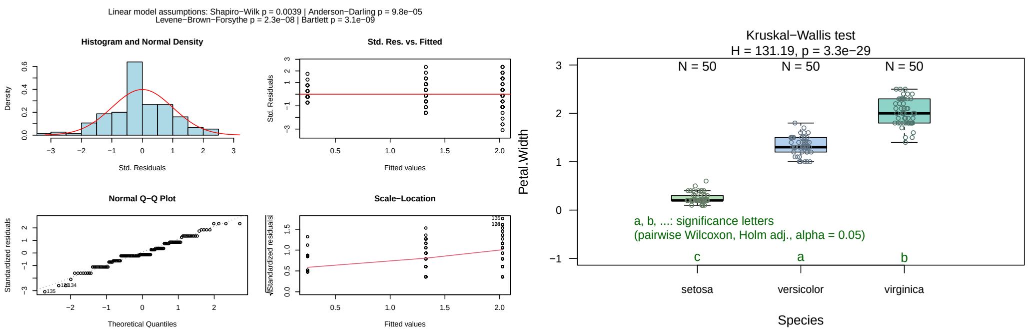

Figure 6: Kruskal-Wallis test applied to the `iris` dataset (`Species` vs. `Petal.Width`). Assumption diagnostics (Shapiro–Wilk rejected) and box plots with the Kruskal-Wallis result; all pairwise post-hoc comparisons significant (all letters differ).

4.3 Both variables numeric

Simple linear regression

`When both predictor and response are numerical, visstat()` checks the standardised residuals from `lm()` graphically and with the Shapiro–Wilk and Anderson–Darling tests; the Breusch–Pagan test assesses homoscedasticity. Regardless of these diagnostics, a simple linear regression fitted and displayed. The graphical shows the fitted line $\hat{y} = b_0 + b_1 x$, where b_0 is the intercept and b_1 the slope, with pointwise confidence and prediction bands at the specified `conf.level`.

The title shows R^2, the estimated regression parameters with confidence intervals and p-values. The returned object contains the regression statistics, the residual-normality p-values, and pointwise estimates of the confidence and prediction bands.

The `swiss` dataset records standardised fertility and socioeconomic indicators for 47 French-speaking Swiss provinces in 1888. We examine how the share of draftees achieving the highest army examination score (`Examination`) predicts the fertility measure (`Fertility`), with `conf.level = 0.99`. Both normality tests pass and the Breusch–Pagan test confirms homoscedasticity, validating the linear model.

```
linreg_swiss <- visstat(swiss$Examination, swiss$Fertility, conf.level = 0.99)
```

Spearman rank correlation

For numeric–numeric input, correlation analysis is never triggered automatically; it requires the explicit flag `correlation = TRUE`. With wind speed (`Wind`) and ozone concentration (`Ozone`) from the `airquality` dataset (Chambers, 2018), a default `visstat()` call fits simple linear regression but flags violated normality and homoscedasticity. The warning recommends two paths: rerun with `correlation = TRUE` within `visstat()`, or switch to a model outside it. Staying within `visstat()`, `correlation = TRUE` gives an assumption-free Spearman result at the cost of causal interpretability.

```
spearman_air <- visstat(airquality$Wind, airquality$Ozone, correlation = TRUE)
```

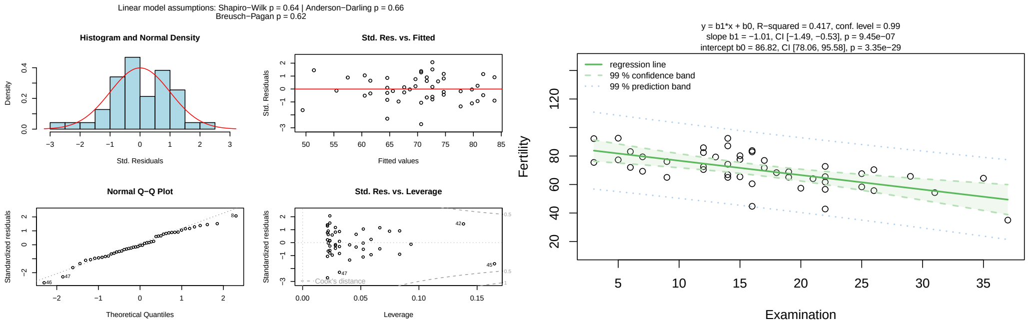

Figure 7: Simple linear regression of `Fertility` on `Examination` for the `swiss` dataset (`conf.level = 0.99`). Assumption diagnostics (Shapiro–Wilk, Anderson–Darling, Breusch–Pagan) and scatter plot with fitted regression line, 99% confidence band (dark shading), and 99% prediction band (light shading).

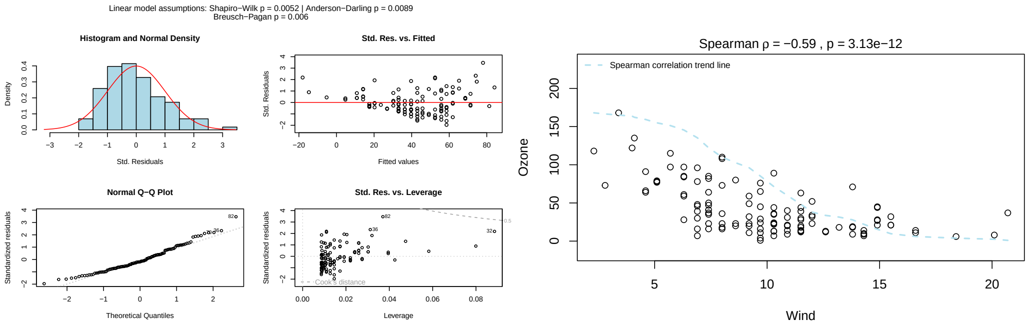

Figure 8: Spearman rank correlation of `Wind` and `Ozone` from the `airquality` dataset (`correlation = TRUE`). Assumption diagnostics and scatter plot of ranked values annotated with ρ and the p-value.

4.4 Both variables categorical

Observed frequencies are arranged in a contingency table, where rows index the levels of the response and columns index the levels of the predictor. `visstat()` tests the null hypothesis that the variables are independent. If expected frequencies satisfy Cochran's rule, Pearson's χ^2 test is used; otherwise, `visstat()` switches to Fisher's exact test. For 2×2 tables, Yates' continuity correction is applied by default to Pearson's χ^2 test. The graphical output depends on the selected test. General $R \times C$ Pearson χ^2 tests show a grouped column plot of row percentages with the p-value in the title, followed by a mosaic plot with colour-coded Pearson residuals. Yates-corrected 2×2 χ^2 tests show the grouped column plot only, because the corrected statistic is not decomposed into cell-level residuals. Fisher's exact test shows absolute counts with N labels above each bar and the p-value in the title.

The following examples are based on the `HairEyeColor` contingency table, which is converted to the column-based data frame expected by `visstat()` using `counts_to_cases()`.

Pearson's χ^2 test

With `Eye` and `Hair` from `HairEyeColor`, all expected cell counts exceed the Cochran thresholds (Cochran, 1954), so the χ^2 approximation is used. For this 4×4 table, `visstat()` returns both a grouped bar chart of row percentages and a mosaic plot with tiles coloured by Pearson residuals. The mosaic plot shows which cells drive the association: blue tiles indicate observed counts above expectation and red tiles indicate observed counts below expectation.

```
hair_eye_df <- counts_to_cases(as.data.frame(HairEyeColor))
visstat(hair_eye_df$Eye, hair_eye_df$Hair)
```

Fisher's exact test

Restricting `HairEyeColor` to male participants with black or brown hair and hazel or green eyes yields a 2×2 table where one expected frequency is less than 5, violating Cochran's rule (Cochran, 1954).

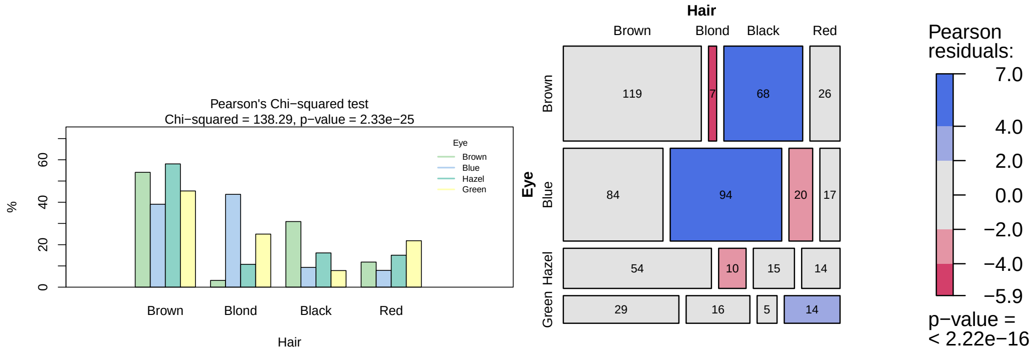

Figure 9: Pearson's χ^2 test applied to the `HairEyeColor` dataset. Grouped bar chart of eye colour by hair colour and mosaic plot with tiles coloured by Pearson residuals (blue: over-represented, red: under-represented).

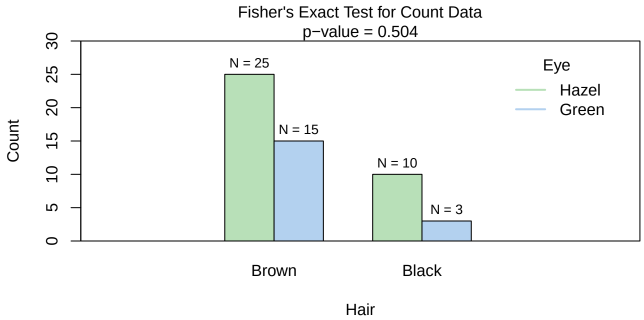

Figure 10: Fisher's exact test applied to a 2×2 subset of `HairEyeColor` (male participants, black/brown hair, hazel/green eyes). Bar chart of absolute counts with the p-value; the odds ratio and confidence interval are accessible from the returned object.

`visstat()` therefore applies Fisher's exact test. For 2×2 tables, the returned object also contains the conditional maximum likelihood estimate of the odds ratio and its confidence interval.

```
hair_eye_male <- HairEyeColor[, , 1]
black_brown_hazel_green <- hair_eye_male[1:2, 3:4]
black_brown_hazel_green_df <- counts_to_cases(
  as.data.frame(black_brown_hazel_green))
fisher_stats <- visstat(black_brown_hazel_green_df$Eye,
                        black_brown_hazel_green_df$Hair)
fisher_stats$estimate   # odds ratio

#> odds ratio
#>  0.5062015

fisher_stats$conf.int   # 95 % CI

#> [1] 0.07725895 2.40885255
#> attr(,"conf.level")
#> [1] 0.95
```

4.5 Ordered factors

When the response is ordered and the predictor is categorical, `visstat()` converts the response to integer level codes and redirects to the non-parametric Wilcoxon or Kruskal–Wallis path above. When both variables are ordered factors and `correlation = TRUE`, `visstat()` instead tests for a monotone association via Kendall's τ_b (Kendall, 1945; Agresti, 2010). The Kendall output is a single jittered rank–rank scatter plot that visualises the monotone trend, colour-codes points by the predictor level, and annotates the plot with τ_b and the p-value.

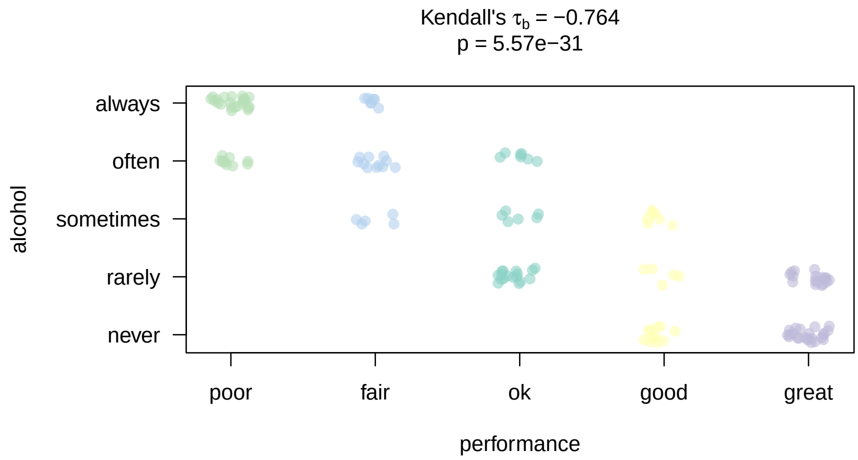

Figure 11: Kendall's τ_b for a hypothetical survey ($n = 150$): alcohol consumption frequency vs. academic performance (both five-point ordinal scales). Jittered rank–rank scatter annotated with τ_b and the p-value; the negative monotone association is significant.

Kendall's τ_b

A hypothetical survey of 150 secondary-school students records alcohol consumption frequency and academic performance on five-point ordinal scales. A negative monotone association is induced by construction: students who consume alcohol more frequently tend to have lower academic performance.

```
set.seed(42)
n <- 150
xs <- sample(1:5, n, replace = TRUE)
ys <- pmin(5, pmax(1, (6 - xs) + sample(-1:1, n, replace = TRUE)))
likert_alc  <- c("never", "rarely", "sometimes", "often", "always")
likert_perf <- c("poor",  "fair",   "ok",        "good",  "great")
alcohol     <- ordered(likert_alc[xs],  levels = likert_alc)
performance <- ordered(likert_perf[ys], levels = likert_perf)
kendall_result <- visstat(performance, alcohol, correlation = TRUE)
```

5 Saving graphical output

All plots can be saved in any format supported by **Cairo** (Urbanek and Horner, 2025) — `"png"`, `"pdf"`, `"svg"`, `"ps"`, `"tiff"` — by supplying the `graphicsoutput` and `plotDirectory` arguments. File names follow the pattern `testname_namey_namex.ext` by default, or a user-supplied `plotName`. The full file paths of generated graphics are stored as the `"plot_paths"` attribute on the returned `"visstat"` object.

6 Limitations

6.1 Scope of automated test selection

All tests are applied with their default settings from base R. As a consequence, paired tests are not supported and interaction terms are not available in the ANOVA path. Only simple linear regression is implemented, as multiple regression cannot be visualised in two dimensions. When regression assumptions are violated, **visStatistics** offers Spearman rank correlation (`correlation = TRUE`) as the sole automated alternative. Alternative methods such as data transformations, generalised linear models, or robust regression are not implemented: each requires user judgment — about the transformation family, the link function, or the estimator — that cannot be automated without substantially expanding the decision tree and increasing the risk of Type I error inflation.

6.2 P-value-based assumption testing

No single test maintains optimal Type I error rates and power across all distributions (Olejnik and Algina, 1987). Normality tests behave poorly at both extremes of the sample-size range: they fail to detect non-normality in small samples and flag negligible departures as significant in large ones (Ghasemi and Zahediasl, 2012; Fagerland, 2012). The $n > 50$ per-group threshold for bypassing the Shapiro–Wilk test is necessarily arbitrary; simulation studies indicate that the required sample size for the central limit theorem to take effect depends on the skewness and kurtosis of the underlying distribution (Fagerland, 2012). Assumption tests provide no information on the nature of the deviation (Shatz, 2024), and combining multiple tests inflates the overall Type I error rate. For these reasons, the diagnostic plots generated by `visstat()` should always be inspected alongside the automated decision; the `conf.level` argument provides a direct handle on the α used throughout.

6.3 Visualisation

The package depends exclusively on base R graphics with no **ggplot2** (Wickham, 2016) dependency, keeping the transitive dependency footprint minimal. For more polished, annotated plots of chosen statistical tests, see **ggstatsplot** (Patil, 2021) or **ggpubr** (Kassambara, 2026).

7 Summary

visStatistics provides a single-function interface to automated statistical test selection, execution, and visualisation for the most commonly used inferential frameworks in two-variable analyses. Its principal contribution is the integration of the full pipeline — data-type detection, distributional assumption checking on GLM residuals, test routing, and annotated graphical output — into a reproducible, server-deployable workflow with a minimal dependency footprint. The package is available on CRAN at `https://CRAN.R-project.org/package=visStatistics` and on GitHub at `https://github.com/shhschilling/visStatistics`.

8 Appendix A: The general linear model

The general linear model (Searle, 1971) provides a unified framework underlying Student's t-test, Fisher's ANOVA, and simple linear regression:

$$Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_{k-1} x_{i,k-1} + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2).$$

Student's t-test uses one binary indicator variable x_{i1}; testing $H_0 : \beta_1 = 0$ is equivalent to testing $H_0 : \mu_1 = \mu_2$.

Fisher's ANOVA uses $k-1$ indicator variables for k groups; testing $H_0 : \beta_1 = \cdots = \beta_{k-1} = 0$ is equivalent to testing equality of all group means.

Simple linear regression uses one continuous predictor; testing $H_0 : \beta_1 = 0$ examines whether a linear relationship exists.

The two-sample case connects t-tests and ANOVA: $t^2 = F$. For two groups, `t.test(var.equal = TRUE)` and `aov()` yield identical p-values, as do `t.test(var.equal = FALSE)` and `oneway.test()`.

9 Appendix B: Test statistics

9.1 Normality tests

Both normality tests are applied to the internally studentised residuals $r_i^* = e_i / (s\sqrt{1-h_{ii}})$, where e_i are the raw residuals, s^2 is the residual mean square, and h_{ii} is the leverage of observation i (Cook and Weisberg, 1982).

Shapiro–Wilk test `shapiro.test()`

The Shapiro–Wilk test (Shapiro and Wilk, 1965) evaluates whether a sample $x_1, \ldots, x_n$ comes from a normal distribution. Let $x_{(i)}$ denote the i-th order statistic. The test statistic is

$$W = \frac{\left(\sum_{i=1}^{n} a_i\, x_{(i)}\right)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2},$$

where $\mathbf{a} = (a_1, \ldots, a_n)^\top$ is the normalised coefficient vector,

$$\mathbf{a} = \frac{V^{-1}\mathbf{m}}{\sqrt{\mathbf{m}^\top V^{-1} V^{-1}\mathbf{m}}},$$

and $Z_{(1)}, \ldots, Z_{(n)}$ are the ordered values expected from a sample of size n drawn from a standard normal distribution. The vector $\mathbf{m}$ contains their expected positions, and V describes how these ordered positions vary together across repeated standard normal samples (Shapiro and Wilk, 1965). W takes values in $(0, 1]$; values close to 1 indicate normality. Simulation studies show that the Shapiro–Wilk test has the highest power among normality tests for small to moderate sample sizes ($n = 10$ to 100) (Razali and Wah, 2011). It drives the parametric/non-parametric branching decision in `visstat()`; for groups with more than 50 observations normality is assumed by the central limit theorem (Lumley et al., 2002; Rasch et al., 2011).

Anderson–Darling test `ad.test()`

The Anderson–Darling test (Anderson and Darling, 1952) is particularly sensitive to deviations in the tails of the distribution (Razali and Wah, 2011; Yap and Sim, 2011). Let $z_i = (x_{(i)} - \bar{x})/s$ be the standardised order statistics and Φ the standard normal CDF. The test statistic is

$$A^2 = -n - \frac{1}{n}\sum_{i=1}^{n}(2i-1)\left[\ln\Phi(z_i) + \ln(1 - \Phi(z_{n+1-i}))\right].$$

The implementation uses `ad.test()` from **nortest** (Gross and Ligges, 2015). The Anderson–Darling result is shown in all assumption plots but does **not** drive the branching decision; only Shapiro–Wilk does.

9.2 Homoscedasticity tests

The Levene–Brown–Forsythe test `levene.test()`

The test (Brown and Forsythe, 1974) improves upon Levene's original proposal (Levene, 1960) by centring on the group median rather than the group mean, making it robust to skewed data and outliers (Allingham and Rayner, 2012). The implementation `levene.test()` mimics the default behaviour of `leveneTest()` in **car** (Fox and Weisberg, 2019) and is the sole driver of the parametric/Welch branching decision.

For each observation y_{ij} in group i, the absolute deviation from the group median $\tilde{y}_i$ is

$$z_{ij} = |y_{ij} - \tilde{y}_i|.$$

The test statistic is the F-statistic from a one-way ANOVA on the z_{ij} values:

$$F = \frac{(N-k)\sum_{i=1}^{k} n_i(\bar{z}_i - \bar{z})^2}{(k-1)\sum_{i=1}^{k}\sum_{j=1}^{n_i}(z_{ij} - \bar{z}_i)^2},$$

where k is the number of groups, $N = \sum_{i=1}^{k} n_i$ is the total sample size, n_i is the sample size of group i, $\bar{z}_i$ is the mean of the absolute deviations within group i, and $\bar{z}$ is the overall mean of all absolute deviations. Under the null hypothesis of equal variances the statistic follows $F(k-1, N-k)$.

Bartlett's test `bartlett.test()`

Bartlett's test (Bartlett, 1937) has greater power than the Brown–Forsythe test when normality holds (Allingham and Rayner, 2012). Its test statistic is

$$K^2 = \frac{(N-k)\ln s_p^2 - \sum_{i=1}^{k}(n_i - 1)\ln s_i^2}{1 + \frac{1}{3(k-1)}\left(\sum_{i=1}^{k}\frac{1}{n_i - 1} - \frac{1}{N-k}\right)},$$

where s_i^2 is the sample variance of group i and s_p^2 is the pooled variance

$$s_p^2 = \frac{1}{N-k}\sum_{i=1}^{k}(n_i - 1)\, s_i^2.$$

Under the null hypothesis the statistic approximately follows $\chi^2(k-1)$. Bartlett's result is shown in the assumption plot but does **not** drive the branching decision; only `levene.test()` does.

Breusch–Pagan test `bp.test()`

For simple linear regression, group-based variance tests are not applicable. The Breusch–Pagan test (Breusch and Pagan, 1979) assesses whether the variance of the residuals from the regression model depends on the predictor values. Its result is displayed in the regression assumption plot.

9.3 Student's t-test and Welch's t-test

The test statistic for Student's t-test (`t.test(var.equal = TRUE)`) is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}},$$

where $\bar{x}_1$ and $\bar{x}_2$ are the sample means, n_1 and n_2 the sample sizes, and s_p the pooled standard deviation

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}},$$

with s_1^2 and s_2^2 the sample variances. The statistic follows a t-distribution with $\nu = n_1 + n_2 - 2$ degrees of freedom.

Welch's t-test relaxes the homoscedasticity assumption. Its statistic is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}},$$

with degrees of freedom approximated by the Welch–Satterthwaite equation (Welch, 1947; Satterthwaite, 1946):

$$\nu \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}.$$

Welch's methods outperform their classical counterparts when variances differ (Moser and Stevens, 1992; Fagerland and Sandvik, 2009; Delacre et al., 2017).

9.4 Wilcoxon rank-sum test

The two-level factor x defines two groups with sample sizes n_1 and n_2. All $N = n_1 + n_2$ observations are pooled and assigned ranks 1 to N. Let $W_1 = \sum_{i=1}^{n_1} R(x_{1,i})$ denote the rank sum of the first group, where $R(x_{1,i})$ is the rank of observation $x_{1,i}$ in the pooled sample. The test statistic returned by `wilcox.test()` (Mann and Whitney, 1947) is the Mann–Whitney U statistic of the first group:

$$W = U_1 = W_1 - \frac{n_1(n_1+1)}{2}.$$

An exact p-value is computed when both groups contain fewer than 50 observations and the data contain no ties; otherwise a normal approximation with continuity correction is used.

9.5 Fisher's one-way ANOVA and Welch's heteroscedastic ANOVA

Fisher's ANOVA test statistic (Fisher and Yates, 1990) is

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}} = \frac{SS_{\text{between}}/(k-1)}{SS_{\text{within}}/(N-k)} = \frac{\sum_{i=1}^{k} n_i(\bar{x}_i - \bar{x})^2 \;/\; (k-1)}{\sum_{i=1}^{k}\sum_{j=1}^{n_i}(x_{ij} - \bar{x}_i)^2 \;/\; (N-k)},$$

where k is the number of groups, $N = \sum_{i=1}^{k} n_i$ is the total sample size, $\bar{x}_i$ is the mean of group i, $\bar{x}$ is the overall mean, and x_{ij} is observation j in group i. Under the null hypothesis $F \sim F(k-1, N-k)$. Post-hoc comparisons use `TukeyHSD()` (Hochberg and Tamhane, 1987), which controls the family-wise error rate via the studentised range distribution (Abdi, 2007).

Welch's heteroscedastic ANOVA (`oneway.test()`) (Welch, 1951) down-weights groups with large variance. Its test statistic is

$$F_W = \frac{\sum_{i=1}^{k} w_i(\bar{y}_i - \bar{y}_w)^2 \;/\; (k-1)}{1 + \frac{2(k-2)}{k^2-1}\sum_{i=1}^{k}\frac{(1 - w_i/w)^2}{n_i - 1}},$$

where $w_i = n_i/s_i^2$ are the inverse-variance weights, $w = \sum_{i=1}^{k} w_i$, and $\bar{y}_w = \sum_{i=1}^{k} w_i\bar{y}_i/w$ is the weighted grand mean. Degrees of freedom are adjusted via a Satterthwaite-type approximation (Satterthwaite, 1946). Post-hoc comparisons use `games.howell()` (Games and Howell, 1976), which applies separate variance estimates per pair. Compact letter displays are produced by `multcompLetters()` from **multcompView** (Graves et al., 2026).

9.6 Kruskal–Wallis test

When normality is rejected, `kruskal.test()` (Kruskal and Wallis, 1952) tests whether the groups come from the same population based on ranks:

$$H = \frac{12}{N(N+1)}\sum_{i=1}^{k} n_i\left(\bar{R}_i - \bar{R}\right)^2,$$

where n_i is the sample size of group i, k is the number of groups, $\bar{R}_i$ is the average rank of group i, N is the total sample size, and $\bar{R} = (N+1)/2$ is the expected average rank under the null hypothesis. The statistic approximately follows $\chi^2(k-1)$. Post-hoc comparisons use Holm-adjusted `pairwise.wilcox.test()` (Holm, 1979).

9.7 Pearson's χ^2 test and Fisher's exact test

Let O_{ij} and E_{ij} denote the observed and expected frequencies in row i and column j of an $R \times C$ contingency table, where rows index the R levels of the response y and columns the C levels of the predictor x. The Pearson residual for cell (i, j) is

$$r_{ij} = \frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}}}, \quad i = 1, \ldots, R, \quad j = 1, \ldots, C.$$

The test statistic of Pearson's χ^2 test (Pearson, 1900) is

$$\chi^2 = \sum_{i=1}^{R}\sum_{j=1}^{C} r_{ij}^2 = \sum_{i=1}^{R}\sum_{j=1}^{C}\frac{(O_{ij} - E_{ij})^2}{E_{ij}},$$

compared to $\chi^2((R-1)(C-1))$. For 2×2 tables, Yates' continuity correction (Yates, 1934) is applied by default. For general $R \times C$ tables, `visstat()` supplements the bar chart with a mosaic plot in which tiles are coloured by r_{ij} (blue: positive, red: negative).

Fisher's exact test (Fisher, 1970) is applied when Cochran's rule (Cochran, 1954) is violated. It evaluates the exact hypergeometric probability of the observed table given the fixed margins. For 2×2 tables with row totals $n_{i\cdot}$ and column totals $n_{\cdot j}$, this probability is

$$\mathbb{P}(N \mid n_{1\cdot}, n_{2\cdot}, n_{\cdot 1}, n_{\cdot 2}) = \frac{\binom{n_{1\cdot}}{n_{11}}\binom{n_{2\cdot}}{n_{21}}}{\binom{n}{n_{\cdot 1}}},$$

where $n = n_{1\cdot} + n_{2\cdot}$ is the total sample size. For 2×2 tables, `fisher.test()` additionally returns the conditional maximum likelihood estimate of the odds ratio $\widehat{\mathrm{OR}} = n_{11}n_{22}/(n_{12}n_{21})$ and its confidence interval.

9.8 Spearman rank correlation

For two numeric variables with `correlation = TRUE`, `visstat()` calls `cor.test(x, y, method = "spearman")` to measure the monotonic association between x and y using ranks, evaluating linear dependence on the ranked data rather than on the original scale. Spearman's ρ is Pearson's r applied to the ranks:

$$\rho = r(\mathrm{rank}(x), \mathrm{rank}(y)) = \frac{\sum_{i=1}^{n}(\mathrm{rank}(x_i) - \overline{\mathrm{rank}(x)})(\mathrm{rank}(y_i) - \overline{\mathrm{rank}(y)})}{\sqrt{\sum_{i=1}^{n}(\mathrm{rank}(x_i) - \overline{\mathrm{rank}(x)})^2}\sqrt{\sum_{i=1}^{n}(\mathrm{rank}(y_i) - \overline{\mathrm{rank}(y)})^2}}.$$

For inference, `cor.test(..., method = "spearman")` computes an exact p-value for small samples without ties by evaluating all $n!$ rank permutations. For larger samples or when ties are present, it uses an approximation to the null distribution of the test statistic. No distributional assumptions on the original data are required.

9.9 Kendall's τ_b

For two ordinal variables with n joint observations, let C denote the number of concordant pairs and D the number of discordant pairs. Kendall's τ_b (Kendall, 1945) is

$$\tau_b = \frac{C - D}{\sqrt{(n_0 - n_1)(n_0 - n_2)}},$$

where $n_0 = n(n-1)/2$, $n_1 = \sum_i t_i(t_i - 1)/2$ sums over groups of tied ranks in the response (with t_i the size of the i-th tie group), and n_2 is the analogous quantity for the predictor. The denominator correction ensures $\tau_b \in [-1, 1]$ even in the presence of ties, which Spearman's ρ does not guarantee (Kendall, 1945). τ_b is therefore preferred for ordinal data with few levels, where ties are unavoidable (Agresti, 2010; Xu et al., 2013).

References

H. Abdi. The Bonferonni and Šidák Corrections for Multiple Comparisons. *Encyclopedia of measurement and statistics*, 2007. [p5, 13]

A. Agresti. *Analysis of Ordinal Categorical Data*. Wiley Series in Probability and Statistics. Wiley, 1 edition, Mar. 2010. ISBN 978-0-470-08289-8 978-0-470-59400-1. doi: 10.1002/9780470594001. [p3, 8, 14]

D. Allingham and J. C. W. Rayner. Testing Equality of Variances for Multiple Univariate Normal Populations. *Journal of Statistical Theory and Practice*, 6(3):524–535, Sept. 2012. ISSN 1559-8608, 1559-8616. doi: 10.1080/15598608.2012.695703. [p11]

T. W. Anderson and D. A. Darling. Asymptotic theory of certain "goodness of fit" criteria based on stochastic processes. *The Annals of Mathematical Statistics*, 23(2):193–212, 1952. doi: 10.1214/aoms/1177729437. [p11]

M. S. Bartlett. Properties of sufficiency and statistical tests. *Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences*, 160(901):268–282, 1937. doi: 10.1098/rspa.1937.0109. [p11]

T. S. Breusch and A. R. Pagan. A Simple Test for Heteroscedasticity and Random Coefficient Variation. *Econometrica*, 47(5):1287–1294, 1979. ISSN 0012-9682. doi: 10.2307/1911963. [p12]

M. B. Brown and A. B. Forsythe. Robust Tests for the Equality of Variances. *Journal of the American Statistical Association*, 69(346):364–367, June 1974. ISSN 0162-1459, 1537-274X. doi: 10.1080/01621459.1974.10482955. [p3, 11]

J. M. Chambers. *Graphical Methods for Data Analysis*. Chapman and Hall/CRC, Boca Raton, Jan. 2018. ISBN 978-1-351-07230-4. doi: 10.1201/9781351072304. [p6]

D. Chicco, A. Sichenze, and G. Jurman. A simple guide to the use of Student's t-test, Mann-Whitney U test, Chi-squared test, and Kruskal-Wallis test in biostatistics. *BioData Min*, 18(1):56, Aug. 2025. ISSN 1756-0381. doi: 10.1186/s13040-025-00465-6. [p1]

W. G. Cochran. The Combination of Estimates from Different Experiments. *Biometrics*, 10(1):101, Mar. 1954. ISSN 0006341X. doi: 10.2307/3001666. [p3, 7, 14]

R. D. Cook and S. Weisberg. *Residuals and Influence in Regression*. New York: Chapman and Hall, 1982. ISBN 0-412-24280-0. [p3, 10]

M. Delacre, D. Lakens, and C. Leys. Why psychologists should by default use Welch's t-test instead of Student's t-test. *International Review of Social Psychology*, 30(1):92–101, 2017. doi: 10.5334/irsp.82. [p12]

M. W. Fagerland. T-tests, non-parametric tests, and large studies—a paradox of statistical practice? *BMC Medical Research Methodology*, 12(1):78, June 2012. ISSN 1471-2288. doi: 10.1186/1471-2288-12-78. [p10]

M. W. Fagerland and L. Sandvik. Performance of five two-sample location tests for skewed distributions with unequal variances. *Contemporary Clinical Trials*, 30(5):490–496, Sept. 2009. ISSN 1551-7144. doi: 10.1016/j.cct.2009.06.007. [p12]

R. A. Fisher. *Statistical Methods for Research Workers*. Oliver and Boyd, Edinburgh, 14th ed., revised and enlarged edition, 1970. ISBN 978-0-05-002170-5. [p14]

R. A. Fisher and F. Yates. *Statistical Methods, Experimental Design, and Scientific Inference: A Re-issue of Statistical Methods for Research Workers, the Design of Experiments and Statistical Methods and Scientific Inference*. Oxford University PressOxford, Apr. 1990. ISBN 978-0-19-852229-4 978-1-383-02363-3. doi: 10.1093/oso/9780198522294.001.0001. [p3, 13]

J. Fox and S. Weisberg. *An R Companion to Applied Regression*. Sage, Thousand Oaks CA, 3 edition, 2019. [p11]

P. A. Games and J. F. Howell. Pairwise multiple comparison procedures with unequal N's and/or variances: A Monte Carlo study. *Journal of Educational Statistics*, 1(2):113–125, 1976. ISSN 0362-9791. doi: 10.2307/1164979. [p3, 13]

A. Ghasemi and S. Zahediasl. Normality Tests for Statistical Analysis: A Guide for Non-Statisticians. *Int J Endocrinol Metab*, 10(2):486–489, Dec. 2012. ISSN 1726-913X, 1726-9148. doi: 10.5812/ijem.3505. [p10]

S. Graves, H.-P. Piepho, and L. S. with help from Sundar Dorai-Raj. *multcompView: Visualizations of Paired Comparisons*, 2026. [p13]

J. Gross and U. Ligges. *Nortest: Tests for Normality*, 2015. [p11]

Y. Hochberg and A. C. Tamhane. *Multiple Comparison Procedures*. Wiley Series in Probability and Statistics. Wiley, 1 edition, Sept. 1987. ISBN 978-0-471-82222-6 978-0-470-31667-2. doi: 10.1002/9780470316672. [p3, 13]

S. Holm. A Simple Sequentially Rejective Multiple Test Procedure. *Scandinavian Journal of Statistics*, 6 (2):65–70, 1979. ISSN 0303-6898. [p3, 13]

A. Kassambara. *Rstatix: Pipe-friendly Framework for Basic Statistical Tests*, 2025. [p1]

A. Kassambara. *Ggpubr: 'ggplot2' Based Publication Ready Plots*, 2026. [p1, 10]

M. G. Kendall. The Treatment of Ties in Ranking Problems. *Biometrika*, 33(3):239–251, 1945. ISSN 0006-3444. doi: 10.2307/2332303. [p3, 8, 14]

W. H. Kruskal and W. A. Wallis. Use of Ranks in One-Criterion Variance Analysis. *Journal of the American Statistical Association*, 47(260):583–621, 1952. ISSN 0162-1459. doi: 10.2307/2280779. [p3, 13]

H. Levene. Robust tests for equality of variances. In I. Olkin, editor, *Contributions to Probability and Statistics: Essays in Honor of Harold Hotelling*, pages 278–292. Stanford University Press, Stanford, CA, 1960. [p11]

T. Lumley, P. Diehr, S. Emerson, and L. Chen. The Importance of the Normality Assumption in Large Public Health Data Sets. *Annu. Rev. Public Health*, 23(1):151–169, May 2002. ISSN 0163-7525, 1545-2093. doi: 10.1146/annurev.publhealth.23.100901.140546. [p1, 3, 11]

H. B. Mann and D. R. Whitney. On a test of whether one of two random variables is stochastically larger than the other. *The Annals of Mathematical Statistics*, 18(1):50–60, 1947. doi: 10.1214/aoms/1177730491. [p3, 12]

D. Meyer, A. Zeileis, and K. Hornik. The strucplot framework: Visualizing multi-way contingency tables with vcd. *Journal of Statistical Software*, 17(3):1–48, 2006. doi: 10.18637/jss.v017.i03. [p3]

D. Meyer, A. Zeileis, K. Hornik, and M. Friendly. *vcd: Visualizing Categorical Data*, 2024. [p3]

B. K. Moser and G. R. Stevens. Homogeneity of variance in the two-sample means test. *The American Statistician*, pages 19–21, Feb. 1992. doi: 10.1080/00031305.1992.10475839. [p12]

S. F. Olejnik and J. Algina. Type I Error Rates and Power Estimates of Selected Parametric and Nonparametric Tests of Scale. *Journal of Educational Statistics*, 12(1):45, 1987. ISSN 03629791. doi: 10.2307/1164627. [p10]

I. Patil. Visualizations with statistical details: The 'ggstatsplot' approach. *Journal of Open Source Software*, 6(61):3167, May 2021. ISSN 2475-9066. doi: 10.21105/joss.03167. [p1, 10]

K. Pearson. On the criterion that a given system of deviations from the probable in the case of a correlated system of variables is such that it can be reasonably supposed to have arisen from random sampling. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 50(302): 157–175, July 1900. ISSN 1941-5982, 1941-5990. doi: 10.1080/14786440009463897. [p13]

D. Rasch, K. D. Kubinger, and K. Moder. The two-sample t test: Pre-testing its assumptions does not pay off. *Stat Papers*, 52(1):219–231, Feb. 2011. ISSN 0932-5026, 1613-9798. doi: 10.1007/s00362-009-0224-x. [p3, 11]

N. M. Razali and Y. B. Wah. Power comparisons of Shapiro-Wilk, Kolmogorov-Smirnov, Lilliefors and Anderson-Darling tests. *Journal of Statistical Modeling and Analytics*, 2(1):21–33, 2011. [p11]

Y. Sato, M. Gosho, K. Nagashima, S. Takahashi, J. H. Ware, and N. M. Laird. Statistical methods in the ¡i¿journal¡/i¿ — an update. *New England Journal of Medicine*, 376(11):1086–1087, 2017. doi: 10.1056/NEJMc1616211. [p1]

F. E. Satterthwaite. An Approximate Distribution of Estimates of Variance Components. *Biometrics Bulletin*, 2(6):110–114, 1946. ISSN 0099-4987. doi: 10.2307/3002019. [p3, 12, 13]

S. R. Searle. *Linear Models*. John Wiley & Sons, 1971. [p1, 10]

S. S. Shapiro and M. B. Wilk. An analysis of variance test for normality (complete samples)†. *Biometrika*, 52(3-4):591–611, Dec. 1965. ISSN 0006-3444. doi: 10.1093/biomet/52.3-4.591. [p3, 10, 11]

I. Shatz. Assumption-checking rather than (just) testing: The importance of visualization and effect size in statistical diagnostics. *Behav Res*, 56(2):826–845, Feb. 2024. ISSN 1554-3528. doi: 10.3758/s13428-023-02072-x. [p10]

I. Subirana, H. Sanz, and J. Vila. Building bivariate tables: The compareGroups package for R. *Journal of Statistical Software*, 57(12):1–16, 2014. [p1]

S. Urbanek and J. Horner. *Cairo: R Graphics Device Using Cairo Graphics Library for Creating High-Quality Bitmap (PNG, JPEG, TIFF), Vector (PDF, SVG, PostScript) and Display (X11 and Win32) Output*, 2025. [p2, 9]

B. L. Welch. The generalization of 'student's' problem when several different population variances are involved. *Biometrika*, 34(1–2):28–35, 1947. doi: 10.1093/biomet/34.1-2.28. [p3, 12]

B. L. Welch. On the Comparison of Several Mean Values: An Alternative Approach. *Biometrika*, 38 (3/4):330–336, 1951. ISSN 0006-3444. doi: 10.2307/2332579. [p3, 13]

H. Wickham. *Ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag New York, 2016. ISBN 978-3-319-24277-4. [p1, 10]

W. Xu, Y. Hou, Y. Hung, and Y. Zou. A comparative analysis of Spearman's rho and Kendall's tau in normal and contaminated normal models. *Signal Processing*, 93(1):261–276, Jan. 2013. ISSN 01651684. doi: 10.1016/j.sigpro.2012.08.005. [p3, 14]

B. W. Yap and C. H. Sim. Comparisons of various types of normality tests. *Journal of Statistical Computation and Simulation*, 81(12):2141–2155, Dec. 2011. ISSN 0094-9655, 1563-5163. doi: 10.1080/00949655.2010.520163. [p11]

F. Yates. Contingency Tables Involving Small Numbers and the $\chi 2$ Test. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 1(2):217–235, July 1934. ISSN 1369-7412, 1467-9868. doi: 10.2307/2983604. [p3, 13]

Sabine Schilling
Institute of Tourism and Mobility, Lucerne University of Applied Sciences and Arts
6002 Lucerne, Switzerland
https://github.com/shhschilling/visStatistics
ORCiD: 0000-0002-8318-9421
sabineschilling@gmx.ch